Scope and Vision

Here at $oftware ¢orp we aim to provide the best software that can possibly be provided to solve problems and provide solutions in any way applicable. With growing amounts of people own cars, we have begun a project, we have been looking into devising how to create and integrate new software to be utilized in parking garages to create a much smoother experience for customers to pay and reserve parking spaces that not only is highly effective, but it is also safe and reliable.

What makes us different

Our company will prioritize safety above all else and provide features that help ensure individuals and their vehicles are as safe as possible. Companies such as Wayleadr and Ronspot are geared for office spots and employees to book spaces offering features that give live views of available parking spaces and notify when spaces are taken by unauthorized vehicles. Other companies such as Flashparking provide users with numerous available options depending on their location while providing a smooth process for acquiring the parking space. At $oftware ¢orp we plan to incorporate many of these features into our company and expand upon them to ensure a safe environment such as notifications of potential flooding and if any levels (such as the ground level) are at risk, alerts of pedestrian density or emergency vehicles within the vicinity.

SRS

- Project: Safe Parking Garage
- Version: 1.7
- Date 11-5-2026
- 1. Intro
    - 1.1 Purpose: To develop and produce an app to use parking garage software that we integrate into parking garages which provides all users with a smooth experience in acquiring a parking spot. Additionally, ensure safety is upheld for anyone in or around the parking garage.
    - 1.2 Scope: Introduce features such as increased camera capabilities, mobile app features that provide seamless access to acquiring parking spaces, but does not introduce features that affect anything outside of the parking garage directly such as working hours of parking garage staff/security guards.
- 2. General Description
    - Users: Registered customers, security guards, parking staff

- System environment: Include a web-based option allowing for QR code scanning, mobile app, operational on desktop
- Constraints: Must be able to run on mobile devices, laptops and desktops
- Assumptions: internet access is available, user has a registered code

- 3. Functional Requirements:
  - FR1: The system allows users to access and acquire parking spaces securely and quickly
  - FR2: The system allows users to view a map displaying parking spots that are available or being taken live
  - FR3: The system allows users to display alerts of high traffic and or pedestrian density in or around the parking garage
  - FR4: The system should automatically create a bar code to be scanned when they arrive at the gates of the parking garage upon completion of parking spot reservation process
  - FR5: The system should notify users of any emergencies (inclement weather, flood warnings, heavy storms or tornado warnings, etc.) that could or will potentially affect parking and which garages will be or are most likely to be affected.
  - FR6:
- 4. Non-Functional Requirements
  - The system must support at least 1,200 users at the same time
  - The system must show spot availability in real-time
  - The system should have 99.999% uptime
- 5. Use cases
  1. Scanning license plates
     a. Camera scans vehicles prior to entering the garage
     b. Plates are compared to databases to determine if they are valid or blacklisted.
     c. Plates that are valid are allowed in
     d. Plates that are not are kept outside of the gates, and currently active staff and or law enforcement are alerted of the current location
  2. Acquiring a parking spot

     Precondition: user is already registered with the app or system

     a. User choose any available parking spot
     b. Payment is processed and spot is reserved to respective user
     c. Cameras at the specific spot scan the license plate for confirmation of correct user

Postcondition: user has secured their registered spot

3. User parks in a spot they did not claim
   Precondition: user is already registered with the app or system
   a. User parks in a spot they did not reserve
   b. Camera scans and determines the plate is incorrect for that spot
   c. On-site staff is notified of which spot is currently being incorrectly occupied
   d. User who claimed the spot is notified of when the spot is and is no longer occupied by the wrong user
4. Event notification
   a. App notifies users of current events and gives link the homepage of the event
   b. Displays map of the event and which garage is the closest to it
5. Emergency notification
   a. Alerts users of emergencies near parking garages
   b. Displays reason for the emergency
   c. Displays if the emergency is ongoing or how long it has been since its conclusion
6. Parking spot availability map display
   a. System shows available parking spots in real time
   b. User is able to select any parking space shown on the map that is available at the time seen
7. Pedestrian density notification
   a. Users are notified of high density of people in the vicinity of parking garages
   b. The system displays map showing which areas have the highest densities
8. Traffic density notification
   a. User is notified of high traffic density in the vicinity of parking garages
   b. The system displays map showing which areas have the highest densities
9. Exiting the parking garage
   a. User scans valid code they acquired when reserving their space
   b. System confirms the scanned code
   c. Gate opens and allows user to exit
10. Display of garages in near proximity of each other

    a. System displays garages in close proximity of each other in order of highest to least available parking spaces
    b. User navigates which parking garage they desire
    c. User chooses which parking space of the decided parking garages they want to reserve
11. Display of EV charging station parking spots
    a. System displays parking spaces that are designated for electric vehicles
    b. User decides which of the available parking spaces they want to reserve
12. Display of user reservation history
    a. Registered user navigates to <Parking history> in the mobile app or website
    b. System displays history of user's parking instances and receipts in order of most to least recent
       Postcondition: User is able to navigate through any of their previous parking instances
13. Parking staff logging in for shifts
    a. Registered employees' login to app or website using their relevant credentials
    b. System searches through database to find employee
    c. System confirms the registered employee and logs them in as current on-site staff
14. Map display of directions to the designated parking garage
    a. User registers for desired parking space
    b. System confirms payment and reservation
    c. System displays option for user to use a link that connects to a third-party service for navigation (ex. Google maps) to the respective parking garage
    d. System gives ETA for the user
    e. System alerts user if any high amounts of traffic or pedestrians, or any ongoing emergencies will be met upon arrival to the parking space
15. Customer support access
    a. User navigates to <customer support>
    b. System displays list of options for user to navigate through or select
    c. User selects desired option

d. System responds with displaying relavent information
Postcondition User accesses desired help information

16. Inclement weather alert (example: potential flooding)
    a. System notifies user of potential flooding and which locations are most likely to be affected
    b. User clicks on notification
    c. System displays map of warning showcasing potentially affected areas
17. Inclement weather protocol (example: potential flooding)
    a. System notifies user of flooding and which garages are affected and thus no longer accessible
    b. User clicks notification
    c. System displays map of which locations are affected and which floors are too
    d. If the flooding event prevents the user from getting to their parking spot during the time in which the event is affecting the garage, the user's spot reservation is cancelled, the user is refunded and prompted with the option of selecting and reserving a new parking space
    e. *exceptions* Users that have already parked their car in their reserved spot will still receive the notification and information of the event and affected parking garages, but they will not receive a refund

WBS for Parking App

- 1. Frontend
  - 1.1 Home screen
    - Login/sign-up (one time sign up, but this option will still be available for account switches)
      - Differentiate between customers and employee
      - "Forgot Password" or "Forgot username" options
        - Prompt with choice of inputting email or phone number to receive recovery code
  - 1.2 Search bar for available parking garages
    - Suggestion with supported locations
  - 1.3 Deciding on parking spot
    - Interactable map display of parking spots
    - Confirmation

        - Payment and receipt screen
    - 1.4 History
        - Show receipts containing all information pertaining to respective previous sessions or parking spaces
- 2. Backend
    - 2.1 Authentication
        - Happens during initial sign up or every time the active account is changed
    - 2.2 Accurate display of available and unavailable parking spaces
    - 2.3 Payment options
        - Card, or other electronic options such as paypal
    - 2.4 connection to weather services and capable of receiving and reporting emergency information to user if they impact any parking garage locations that is equipped with the software

Trello Scrum Board- Visualization of all tasks and features needed to be completed and implemented into the parking software

- https://trello.com/b/NQjqDHto/cis-4374-scrum-board

Risks

- Technical
    1. Camera unable to read license plates properly due to raining or thunder storming
    2. App completely not working due to system dysfunctionality
    3. Map capabilities not working properly due to app being unable to connect to 3rd party services (ex. Google Maps)
    4. Parking spot live count not tracking correctly due to system errors
    5. Errors with kiosk during ticket reading
    6. Dysfunctionality with the gate opening
    7. Extreme weather such as extreme heat may impact data centers that would in turn impact the parking garage software performance/functionality
- Financial
    1. Features increase as project develops without budget increases
    2. Task requirements are underestimated and leads to needing more financial backing than what is available
    3. Initial supplier unable to provide required component to develop kiosk, leading to needing to acquire the part from a different supplier that is more expensive

4. Payment not received properly due to transaction systems' dysfunctionality
5. Extreme weather damages kiosk, cameras, or gate leading to requiring repairs/replacement

- Schedule
    1. Task/sprint not completed within deadline
    2. Major inclement weather prevents development on the project
    3. Software integration taking longer than expected to properly implement
    4. Solving potential errors or problems found takes longer than desired leading to task work and completion being pushed back
    5. Task or developments relying on responses of higher-ups or other members
- People
    1. Team member quit from project
    2. Team member overburdened with workload
    3. Team member/group undertrained or inexperienced with certain aspects of the system such as how to operate the functions from an operator side
    4. Burn-out or disappointment from team or stakeholders for any reason such as development progress

Communication Plan

- Teams and Cadence
    - Project Manager and Leads-15-minute-long Daily stand-ups via online meetings discussing immediate tasks or changes
    - Different teams (technical, design, etc.)- hour long weekly meetings discussing progress and make any adjustments to future development if needed
    - Leadership-every 3 weeks, 30-minute updates on total progress, current statuses, and financial situation
    - Stakeholders-30 to 45-minute updates on project progress, end-goals, and any significant changes
- Reporting Methods
    - Microsoft Teams meetings for status updates and daily stand-ups
    - Trello scrum board for visualizations and progress showcases
    - Specific channels directed at relevant people for any urgent notices

Cost Estimation and RACI

RACI Table

Task/Role	Developer	PM	Tester	PO	Dev Ops

Requirements	C	A	I	A	C
Design	R	A	C	C	R
Implementation	R	C	I	C	R
Testing	C	I	R	C	I
Release	I	I	I	I	I

Roles

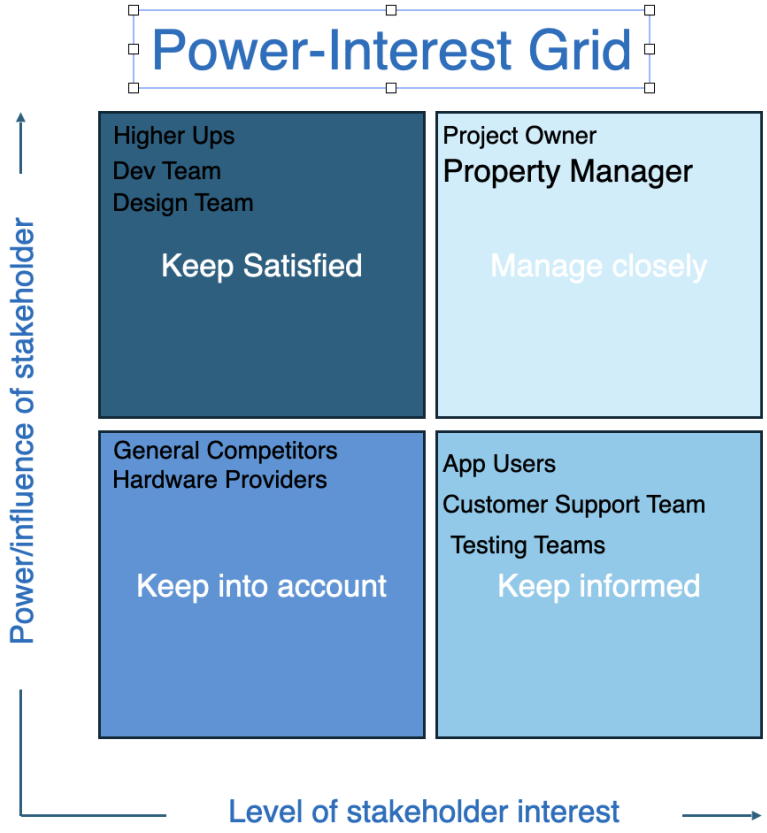
Task/Role	Name	Experience level	FTE %	Estimated hours	Rate/ hr	Total cost $
Project Manager	Branden	Beginner	0.20	250	50	12,500
Developer 1	Marshawn	Senior	0.34	400	60	24,000
Developer 2	Nihyah	Senior	0.33	400	60	24,000
Developer 3	Abigail	Senior	0.33	400	60	24,000
Devops 1	Henrietta	Senior	0.50	400	60	24,000
Devops 2	Adam	Senior	0.50D	400	60	24,000
Tester 1	Mike	Senior	0.50	300	40	12,000
Tester 2	Misty	Senior	0.50	300	40	12,000

Cost Estimation (Bottom up)

- Project manager (1): (250 hours *50/hr) =12,500
- Developers (3): 3 * (400 hours * 60/hr) =72,000
- Devops (2): 2 * (400 hours * 60/hr) =48,000
- Testers (2): 2 * (300 hours * 40/hr) =24,000
- Labor subtotal: 12,500 + 72,000 + 24,000 +48,000 =156,500
- Physical equipment and integration- Cameras and installation with infrared capabilities 90,000; kiosks 20,000; security 85,000; API(s) 16,000, training 15,000
- Subtotal: 156,500 + 90,000 + 20,000 + 85,000 + 16,000 + 15,000 =**382,500**
- Overhead/benefits (20%): 382,500 + (.20 * 382,500) =459,000
- Contingency (12%): 459,000 + (.12 * 459,000) =514,080
- Total: **$514,080**

Team Management

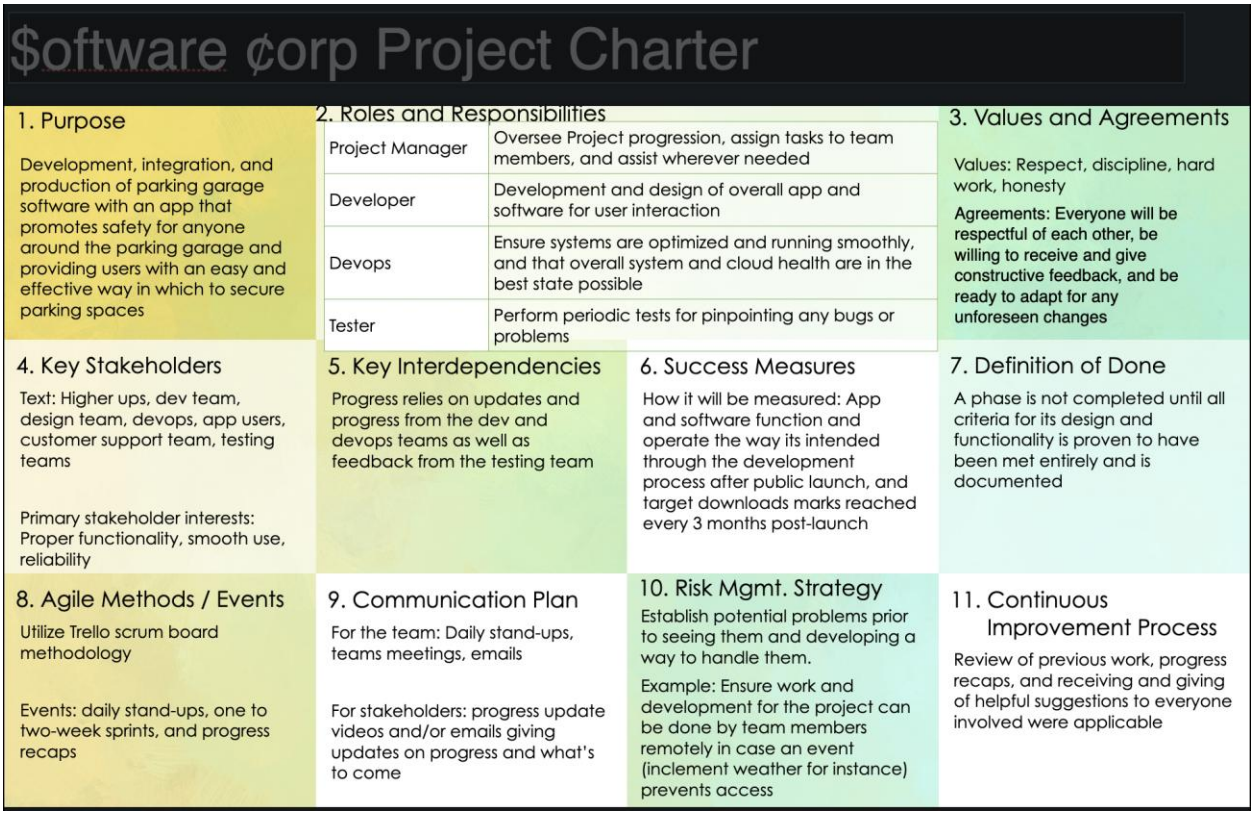
Power Interest Grid



Team Charter



Project Charter



Closure

Executive Summary:

Problem: Develop and implement parking garage software that produces an easy process for reserving and acquiring a parking space. Additionally, ensure that the software and equipment promotes safety for anyone in or around the parking garages

Scope: Introduction of features and camera capabilities that promote seamless use of a parking garage app for both staff and users that ensures minimal effort to use with effective functionality, informs users of various conditions that may affect the route to their parking space, and promotes safety.

Requirements and resulting costs:

- High amounts and constant communication using Microsoft Teams and Trello scrum boards
- Teams of developers, testers, and devops for production, completion, and implementation of the parking garage software
- Trustworthy and effective cloud and APIs
- Final costs came to $520,000 due to unforeseen issues and delays leading to project completion being pushed back slightly past the estimated completion date

Achievements and shortfalls:

- The project reached completion close to the estimated completion date minorly exceeding estimated costs and the delays did not cause the project to be excessively past the estimated completion date.
- Some poor task management and allocation to team members was the fault of the project manager, though the issues were quickly rectified allowing development to quickly get back on schedule shortly after realizing the problem
- Ensuring team members were able to work outside of the main building location ensured inclement weather did not stop development

Lessons learned: The project manager needs to ensure he understands his team members' strengths and weaknesses so that tasks can be allocated in the most effective manner possible as to not allow for any delays or lengthening of work to happen. Though his recovery ensured the issues were not significant, this aspect can be improved to mitigate risk.

Recommendations for future projects:

- Development of software for monitoring flight tickets and creating an easier process for customers to buy and reserve plane seats whether it be one flight or multiple connecting flights
- Healthcare software for monitoring various attributes of patient health especially those in more severe conditions
- Health software for monitoring caloric intake and other attributes of physical health such as body weight and body fat percentage.